

Israel Asefa

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SUMMARY

Backend engineer with 4+ years building high-integrity payment systems, credit ledgers, wallet workflows, and enterprise integrations. Re-architected a client-authoritative credit system into a backend-controlled ledger, eliminating an entire class of support incidents. Strong track record in API design, payment orchestration, transactional database workflows, and production debugging under real business load.

EDUCATION

HiLCoE School of Computer Science

Bachelor of Science in Computer Science

Addis Ababa, Ethiopia

2022

EXPERIENCE

Java Backend Engineer

NiteShift Systems

Feb 2024 – Present

Remote

- Re-architected the credit system from a frontend-authoritative flow into a backend-controlled ledger and synchronization model, eliminating all credit-related support tickets post-release.
- Centralized credit calculation, mutation rules, available-balance checks, and sensitive action validation inside Spring Boot service workflows.
- Orchestrated Stripe checkout, webhook handling, post-payment access updates, subscription state, and purchase-linked content ownership for digital music products.
- Modeled users, releases, purchases, subscriptions, ownership records, credit records, and content metadata using PostgreSQL and Hibernate/JPA.
- Resolved production issues across authentication, payment state, ownership checks, credit logic, admin workflows, release visibility, and content access enforcement.

Software Engineer, Contract Consultant

ETTA Solutions

Jun 2023 – Sep 2024

Addis Ababa, Ethiopia

- Architected Tender Management and Internal Transfer workflows that moved manual business processes into structured approval, tracking, and operational backend systems.
- Delivered an ERCA-compliant document signature and verification service using hashing-based validation to detect document authenticity and tampering.
- Orchestrated Mastercard payment processing, digital wallet transaction flows, SMS notifications, and legacy CNET data migration into business backend systems.
- Designed PostgreSQL schemas for payment workflows, POS data, procurement records, internal transfers, product data, and operational history.
- Resolved production issues across payment, POS, document verification, and internal operations systems under active business use.

Software Engineer

Ahaz Software Solutions

Feb 2021 – Dec 2022

Addis Ababa, Ethiopia

- Automated ZKTeco biometric attendance ingestion using Python workflows that captured, normalized, and persisted device data into HRMS records, removing manual data entry.
- Authored payroll calculation and attendance-processing logic for salary workflows and periodic attendance-based records.
- Designed REST APIs for employee records, payroll processing, attendance records, administrative workflows, and HR reporting.

- Delivered Node.js/Express APIs for the Akila project covering employee attendance, site allocation, and painting site specifications.
- Stabilized HRMS, wallet, and client business systems through debugging, deployment support, and client-facing delivery work.

PROJECTS

Threadline Runtime Evidence & Flow Reconstruction | FastAPI/Python backend for collecting runtime events and reconstructing actual backend execution flows.

- Designed FastAPI ingestion endpoints, Pydantic v2 schemas, and PostgreSQL persistence for raw runtime events emitted by application SDKs.
- Engineered deterministic reconstruction services that order raw events, detect request start/completion, extract errors, summarize database activity, and generate flow case files.
- Containerized local development with Docker and migration-driven database setup for repeatable backend testing.

Odoo Infrastructure Control Plane | Self-hosted Odoo instance management platform for provisioning, configuring, and operating Odoo deployments across remote Linux servers.

- Architected a Spring Boot backend as the control plane and source of truth for host registration, Odoo instance management, deployment actions, and operational state.
- Engineered secure SSH credential handling using encrypted password persistence, allowing deployment services to access remote hosts without storing plaintext credentials.
- Designed a Python provisioning layer to execute deployment actions over SSH, manage Docker-based Odoo/PostgreSQL services, and support start, stop, update, and configuration workflows.

TECHNICAL SKILLS

Backend: Java, Spring Boot, Spring MVC, Spring Data JPA, Hibernate/JPA, Spring Security, JWT, Node.js, FastAPI

Databases: PostgreSQL, MySQL, MongoDB, SQL, relational schema design, transaction modeling

Payments & Integrations: Stripe, Mastercard, wallet systems, webhooks, SMS services, external APIs, biometric device integrations

Architecture & Messaging: REST API design, API contracts, microservices, asynchronous workflows, Apache Kafka

Testing & Delivery: Unit testing, integration testing, Git, code reviews, Agile/Scrum, production support

DevOps: Docker, Jenkins, Kubernetes